Z-Transform

- To use LTI system more easily using transformation method.
- Transform time to frequency domain.
- Z-transform is very useful to analysis discrete system.
- For Continuous system, we use Laplace transform and for discrete system, we use Z-transform.

Two sided or bilateral z-transform

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

Unilateral z-transform

$$X(z) = \sum_{n=0}^{\infty} x[n] z^{-n}$$

- Pole:
  - The *pole* of a z-transform $X(z)$ are the values of z for which $X(z) = \infty$.
- Zero:
  - The *zero* of a z-transform $X(z)$ are the values of z for which $X(z)=0$.
- When $X(z) = P(z)/Q(z)$ is a rational form, and both $P(z)$ and $Q(z)$ are polynomials of z, the poles of are the roots of $Q(z)$, and the zeros are the roots of $P(z)$, respectively.

- Zeros of a system function
  - The system function of the FIR system $y[n] = 6x[n] - 5x[n-1] + x[n-2]$ has been shown as

$$H(z) = 6\frac{\left(z-\frac{1}{3}\right)\left(z-\frac{1}{2}\right)}{z^2} = \frac{P(z)}{Q(z)}$$

- The zeros of this system are $1/3$ and $1/2$, and the pole is 0.
- Since 0 and 0 are double roots of $Q(z)$, the pole is a second-order pole.

- Right-sided sequence:
  - A discrete-time signal is right-sided if it is nonzero only for $n \geq 0$.
- Consider the signal $x[n] = a^n u[n]$.

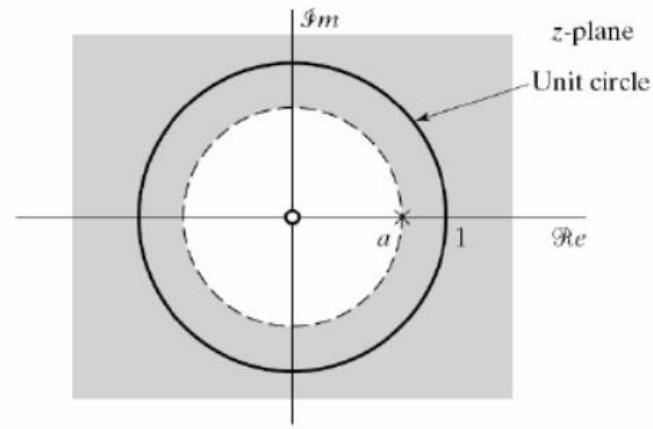
$$X(z) = \sum_{n=-\infty}^{\infty} a^n u[n] z^{-n} = \sum_{n=0}^{\infty} \left(az^{-1}\right)^n$$

- For convergent $X(z)$, we need $\sum_{n=0}^{\infty} \left(az^{-1}\right)^n < \infty$
  - Thus, the ROC is the range of values of z for which $|az^{-1}| < 1$ or, equivalently, $|z| > a$.

- By sum of power series,

$$X(z)=\sum_{n=0}^{\infty}\left(az^{-1}\right)^n=\frac{1}{1-az^{-1}}=\frac{z}{z-a},\quad |z|>|a|$$

- There is one zero, at $z=0$, and one pole, at $z=a$.



○ : zeros

× : poles

Gray region: ROC

- Left-sided sequence:
  - A discrete-time signal is left-sided if it is nonzero only for $n \le -1$.
- Consider the signal $x[n]=-a^n u[-n-1]$.

$$X(z)=-\sum_{n=-\infty}^{\infty}a^n u[-n-1]z^{-n}=-\sum_{n=-\infty}^{-1}a^n z^{-n}$$

$$=-\sum_{n=1}^{\infty}a^{-n}z^n=1-\sum_{n=0}^{\infty}\left(a^{-1}z\right)^n$$

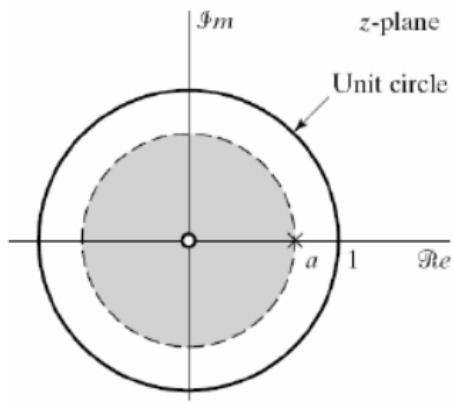
  - If $|az^{-1}|<1$ or, equivalently, $|z|<a$, the sum converges.

$|a^{-1}z|<1$

- By sum of power series,

$$X(z) = 1 - \frac{1}{1-a^{-1}z} = \frac{-a^{-1}z}{1-a^{-1}z} = \frac{z}{z-a}, \quad |z| < |a|$$

- There is one zero, at $z=0$, and one pole, at $z=a$.



The pole-zero plot and the algebraic expression of the system function are the same as those in the previous example, but the ROC is different.

Given $$x(n) = \left(\frac{1}{2}\right)^n u(n) + \left(-\frac{1}{3}\right)^n u(n)$$

Then $$X(z) = \sum_{n=-\infty}^{\infty} \left(\frac{1}{2}\right)^n u(n) z^{-n} + \sum_{n=-\infty}^{\infty} \left(-\frac{1}{3}\right)^n u(n) z^{-n}$$

$$= \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n z^{-n} + \sum_{n=0}^{\infty} \left(-\frac{1}{3}\right)^n z^{-n}$$

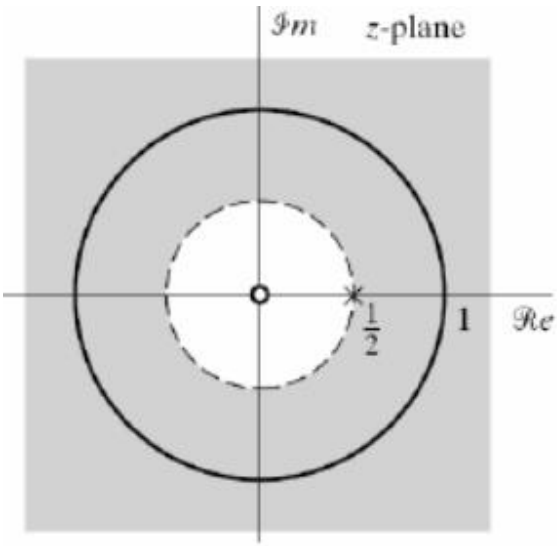
$$= \frac{1}{1-\frac{1}{2}z^{-1}} + \frac{1}{1+\frac{1}{3}z^{-1}} = \frac{2z\left(z-\frac{1}{12}\right)}{\left(z-\frac{1}{2}\right)\left(z+\frac{1}{3}\right)}$$

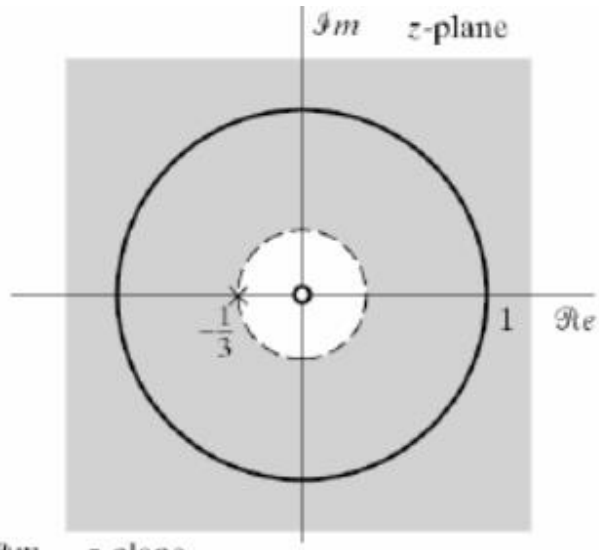
$$\left(\frac{1}{2}\right)^n u(n) \overset{z}{\longleftrightarrow} \frac{1}{1-\frac{1}{2}z^{-1}}, \qquad |z| > \frac{1}{2}$$

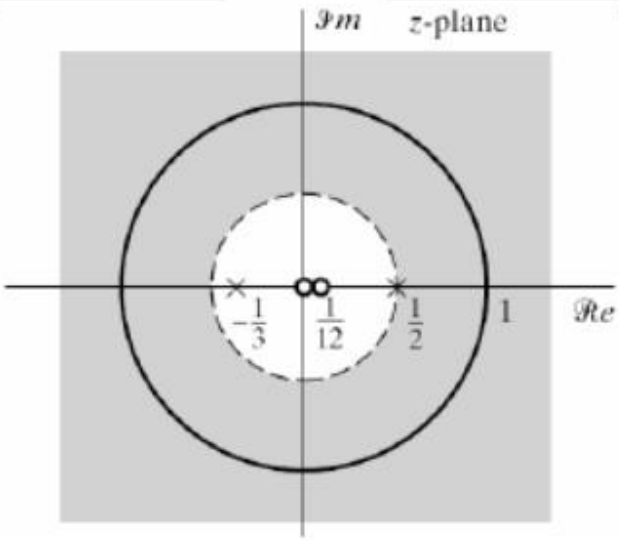
$$\left(-\frac{1}{3}\right)^n u(n) \overset{z}{\longleftrightarrow} \frac{1}{1+\frac{1}{3}z^{-1}}, \qquad |z| > \frac{1}{3}$$

Thus

$$\left(\frac{1}{2}\right)^n u(n) + \left(-\frac{1}{3}\right)^n u(n) \overset{z}{\longleftrightarrow} \frac{1}{1-\frac{1}{2}z^{-1}} + \frac{1}{1+\frac{1}{3}z^{-1}}, \qquad |z| > \frac{1}{2}$$


$\mathcal{I}m$
z-plane
$\frac{1}{2}$
1
$\mathcal{R}e$


$\mathcal{I}m$
z-plane
$-\frac{1}{3}$
1
$\mathcal{R}e$


$\mathcal{I}m$
z-plane
$-\frac{1}{3}$
$\frac{1}{12}$
$\frac{1}{2}$
1
$\mathcal{R}e$

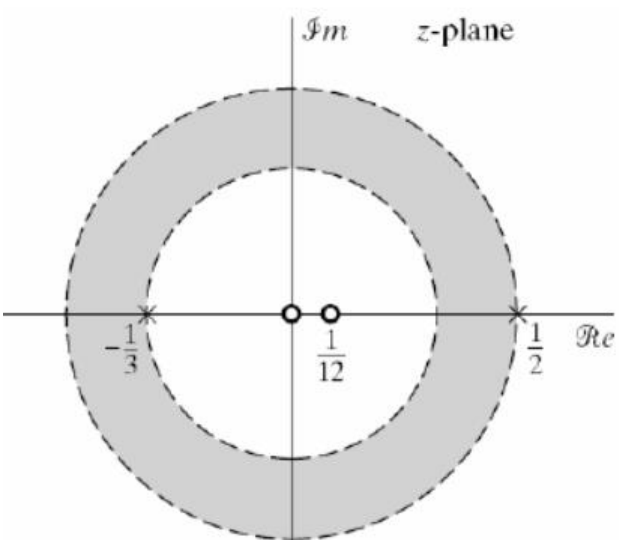
Given $x(n) = \left(-\frac{1}{3}\right)^n u(n) - \left(\frac{1}{2}\right)^n u(-n-1)$

Since $\left(-\frac{1}{3}\right)^n u(n) \overset{z}{\longleftrightarrow} \dfrac{1}{1+\frac{1}{3}z^{-1}}, \qquad |z| > \frac{1}{3}$

and by the left-sided sequence example

$$-\left(\frac{1}{2}\right)^n u(-n-1) \overset{z}{\longleftrightarrow} \frac{1}{1-\frac{1}{2}z^{-1}}, \qquad |z| < \frac{1}{2}$$

$$X(z) = \frac{1}{1+\frac{1}{3}z^{-1}} + \frac{1}{1-\frac{1}{2}z^{-1}} = \frac{2z\left(z-\frac{1}{12}\right)}{\left(z+\frac{1}{3}\right)\left(z-\frac{1}{2}\right)}$$



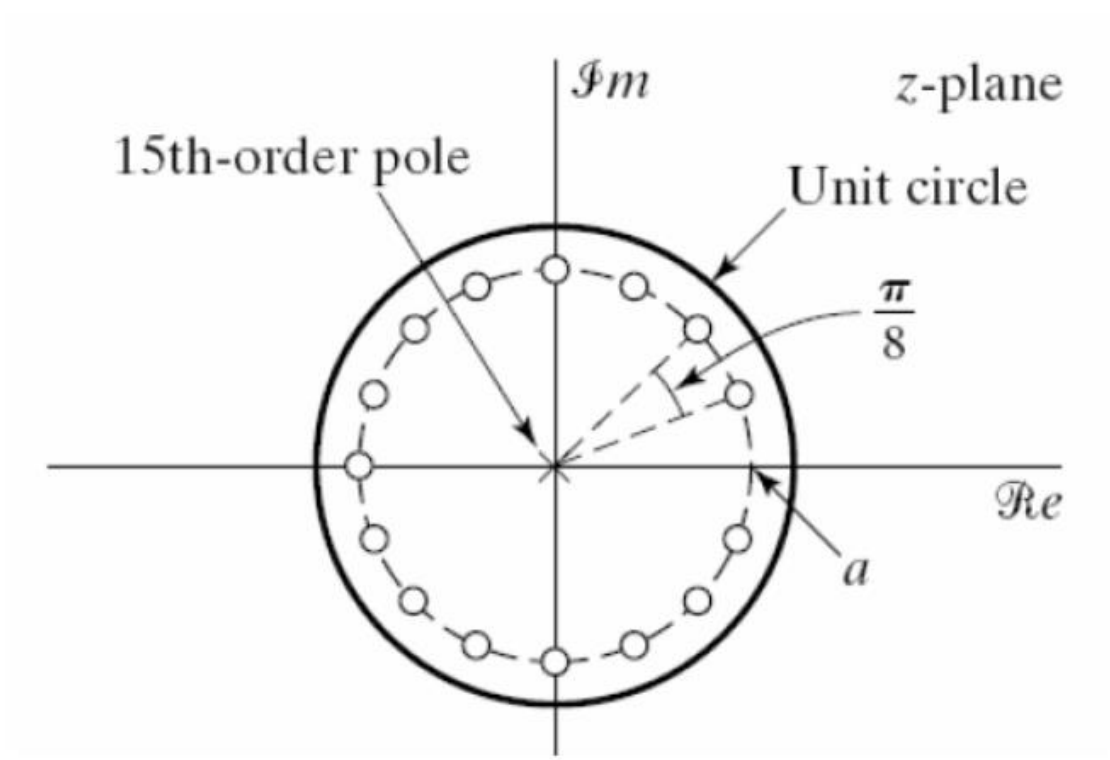
Again, the poles and zeros are the same as the previous example, but the ROC is not.

Given $x(n) = \begin{cases} a^n & 0 \le n \le N-1 \\ 0 & otherwise \end{cases}$

Then $$X(z) = \sum_{n=0}^{N-1} a^n z^{-n} = \sum_{n=0}^{N-1} \left(az^{-1}\right)^n = \frac{1-\left(az^{-1}\right)^N}{1-az^{-1}}$$

$$= \frac{1}{z^{N-1}} \frac{z^N - a^N}{z-a}$$

There are the N roots of $z^N = a^N$, $z_k = ae^{j(2\pi k/N)}$. The root of $k = 0$ cancels the pole at $z=a$. Thus there are $N-1$ zeros, $z_k = ae^{j(2\pi k/N)}$, $k = 1 \ldots N$, and a $(N-1)$th order pole at zero.



$$\delta[n] \leftrightarrow 1 \qquad \text{ROC: all } z.$$

$$u[n] \leftrightarrow \frac{1}{1-z^{-1}} \qquad \text{ROC: } |z| > 1.$$

$$-u[-n-1] \leftrightarrow \frac{1}{1-z^{-1}} \qquad \text{ROC: } |z| < 1.$$

$$\delta[n-m] \leftrightarrow z^{-m} \qquad \text{ROC: all } z \text{ except 0 (if } m > 0) \text{ or } \infty \text{ (if } m < 0).$$

$$a^n u[n] \leftrightarrow \frac{1}{1-az^{-1}} \qquad \text{ROC: } |z| > |a|.$$

$$-a^n u[n] \leftrightarrow \frac{1}{1-az^{-1}} \qquad \text{ROC: } |z| < |a|.$$

$$na^n u[n] \leftrightarrow \frac{az^{-1}}{\left(1-az^{-1}\right)^2} \qquad \text{ROC: } |z| > |a|.$$

$$-na^n u[-n-1] \leftrightarrow \frac{az^{-1}}{\left(1-az^{-1}\right)^2} \qquad \text{ROC: } |z| < |a|.$$

$$[\cos w_0 n]u[n] \leftrightarrow \frac{1-[\cos w_0]z^{-1}}{1-[2\cos w_0]z^{-1}+z^{-2}} \qquad \text{ROC: } |z| > 1.$$

$$[\sin w_0 n]u[n] \leftrightarrow \frac{[\sin w_0]z^{-1}}{1-[2\cos w_0]z^{-1}+z^{-2}} \qquad \text{ROC: } |z| > 1.$$

$$\left[r^n \cos w_0 n\right]u[n] \leftrightarrow \frac{1-[r\cos w_0]z^{-1}}{1-[2r\cos w_0]z^{-1}+r^2z^{-2}} \qquad \text{ROC: } |z| > r.$$

$$\left[r^n \sin w_0 n\right]u[n] \leftrightarrow \frac{[r\sin w_0]z^{-1}}{1-[2r\cos w_0]z^{-1}+r^2z^{-2}} \qquad \text{ROC: } |z| > r.$$

$$\begin{cases} a^n & 0 \le n \le N-1 \\ 0 & otherwise \end{cases} \leftrightarrow \frac{1-a^N z^{-N}}{1-az^{-1}} \qquad \text{ROC: } |z| > 0.$$

Example: $y(n) - y(n-1) + \frac{3}{16} y(n-2) = x(n) - \frac{1}{2} x(n-1)$

<u>Soln.</u> Step 1: Apply Z.T.

$$Y(z) - z^{-1} Y(z) + \frac{3}{16} z^{-2} Y(z) = X(z) - \frac{1}{2} z^{-1} X(z)$$

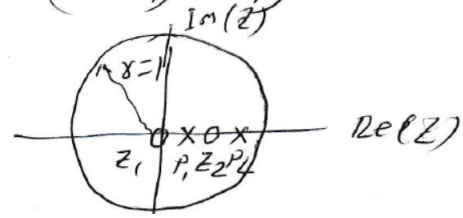
Step 2: $Y(z)\left[1 - z^{-1} + \frac{3}{16} z^{-2}\right] = X(z)\left[1 - \frac{1}{2} z^{-1}\right]$

$$\therefore H(z) = \frac{Y(z)}{X(z)} = \frac{1 - \frac{1}{2} z^{-1}}{1 - z^{-1} + \frac{3}{16} z^{-2}} = \frac{z^2(1 - \frac{1}{2} z^{-1})}{z^2(1 - z^{-1} + \frac{3}{16} z^{-2})}$$

$$= \frac{z^2 - \frac{1}{2} z}{z^2 - z + \frac{3}{16}} = \frac{z(z - \frac{1}{2})}{(z - \frac{1}{4})(z - \frac{3}{4})}$$

Poles: $P_1 = \frac{1}{4}$, $P_2 = \frac{3}{4}$

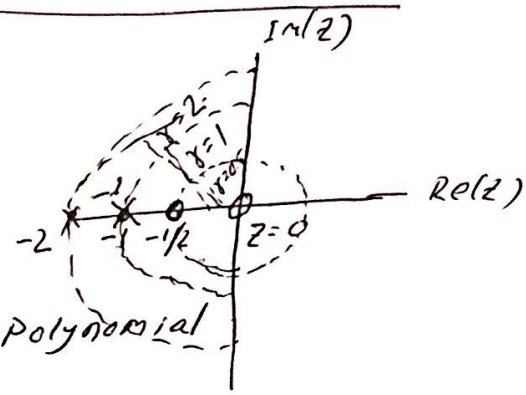
Zeros: $z = 0$, $z = \frac{1}{2}$



Significance of Pole-Zero of Transfer Function:

Let, Transfer Function:

$$H(z) = \frac{z(2z+1)}{(z+1)(z+2)}$$



Zeros are the roots of Numerator Polynomial

Poles are the roots of Denominator

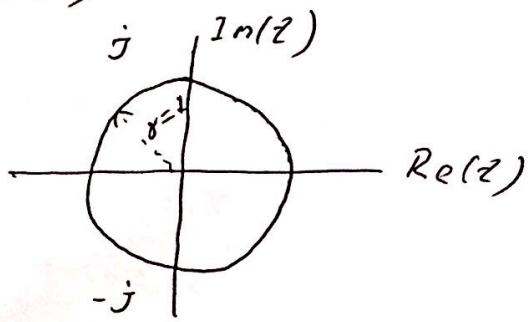
$\therefore\ z(2z+1) = 0 \Rightarrow z = 0$ or $z = -\frac{1}{2}$

$(z+1)(z+2) \Rightarrow z = -1$ or $z = -2$

$e^{+j\pi} = \cos\pi + j\sin\pi = -1 + 0 = -1$

$\frac{1}{2} e^{j\pi}$

Z-plane



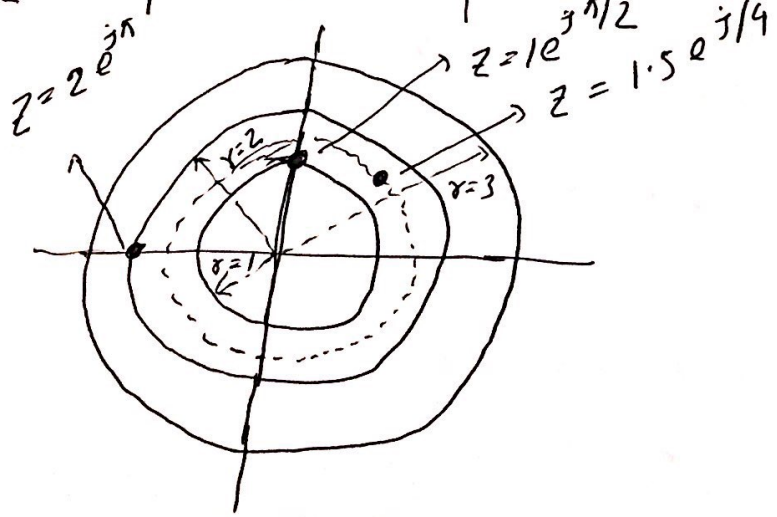
Basic z-plane is written as: $z = r e^{j\theta}$ → angle (phase)

r → radious (Amplitude)

⇒ Polar representation of the complex number (i.e., z is a complex number)

Let, we select: $r = 1$, $r = 2$, $r = 3$

Point:		
$P = 2e^{j\pi}$	$P = 1e^{+j\pi/2}$	$P = 1.5e^{j\pi/4}$
$\theta = \pi$	$\theta = \pi/2$	$\theta = \pi/4$
$r = 2$	$r = 1$	$r = 1.5$



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Zeros make the $H(z)$ to go to zero (0)

Poles " " " " " " (∞)

We know, $H(z) = \frac{Y(z)}{X(z)}$ $\therefore Y(z) = H(z) \cdot X(z)$

$H(z)$ → System Transfer Function

$Y(z)$ → O/P, $H(z)$ → T.F., $X(z)$ → I/P

If T.F is zero then any value of I.P.F is going to ~~O.P.F is~~ zero for O.P.F. (Output Function).

If T.F " ∞ " " " " " " ∞ to O.P.F.

If we want to certain input should be blocked at the output side. then make my $Y(z)$ to go to 0 (zero).

If " " " " " is going to highest amplitude or highest system response then we will keep our $H(z)$ to go to ∞.

Hence, whenever we want ~~our~~ to reject some input then we will put a zeros.

And " " " " pass " " "

" " " " poles.

Poles is representation to pass a certain values and zeros a " to reject " " ".

Poles → Passes the Input
Zeros → Reject " " .

Poles & zeros might be real or complex.
" " " radious and angle.

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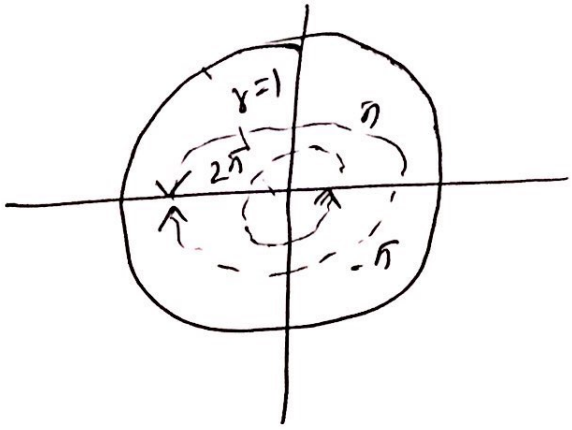
$z = \gamma e^{j\omega}$ ⟶ angular frequency

ω → digital angular frequency, unit is radians.



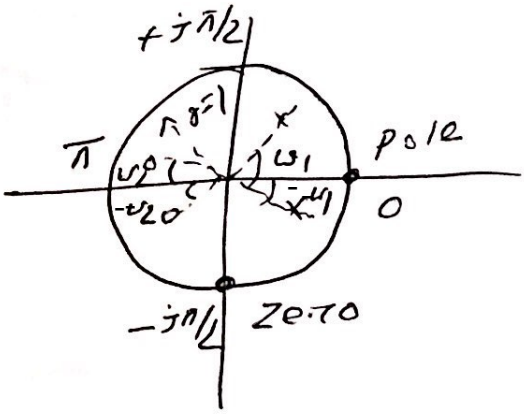
$z = e^{j\omega}$

$\omega = 0 - 2\pi$

$-\pi - \pi$

$z = 1 . e^{-j\pi/2}$

$z_p = 1 . e^{0}$



$\omega = \frac{\pi}{2}$ rejected

$\omega = 0$ Passed

$\omega_1 = -\omega_1$ Passed

$\omega_2 = -\omega_2$ rejected.

Whenever, we wanted reject the ~~pass to~~ frequency then we will put a zero

Whenever, we wanted to pass the frequency the ~~will~~ we will put a pole

In this way, the ~~transf~~ transform function effects the frequency components of the input.

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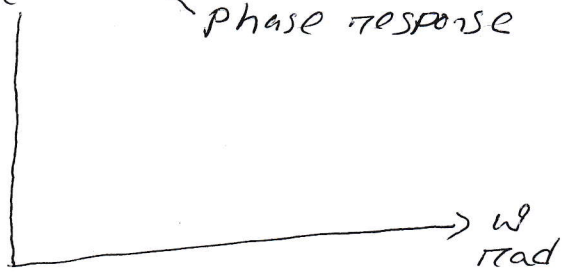
Frequency Response

- It is a complex function which is plotted with varied values of ω (omega).
- We cannot plotted the complex number directly. Therefore, we can transform the complex number to polar formation with magnitude and phase response.

$H(e^{jw})$ — magnitude response
— phase response

$w = 0$ to 2π
or
$-\pi$ to π

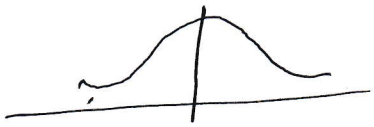


$\therefore$ Frequency response — magnitude response
— phase response

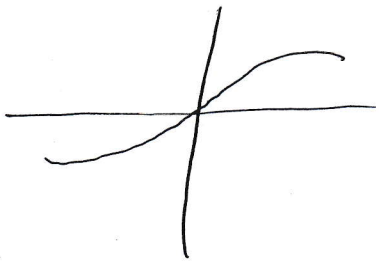
if $h(n) \rightarrow$ real [$h(n)$ is impulse response]

EXP. $h(n) = \{-1, 0, 1, 2, 4\}$

$|H(e^{jw})| \rightarrow$ magnitude response is symmetrical (even)



$\angle H(e^{jw}) \rightarrow$ phase response, is symmetrical (odd)



Example of Frequency Response

Find the magnitude & Phase response for the system.

$$y(n) = \frac{1}{6}x(n) + \frac{1}{3}x(n-1) + \frac{1}{6}x(n-2)$$

Soln: Step 1. Apply Z.T.

$$Y(z) = \frac{1}{6}X(z) + \frac{1}{3}z^{-1}X(z) + \frac{1}{6}z^{-2}X(z)$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1}{6} + \frac{1}{3}z^{-1} + \frac{1}{6}z^{-2}$$

$$\therefore H(z) = \frac{1}{6} + \frac{1}{3}z^{-1} + \frac{1}{6}z^{-2}$$

Step 2: Replace $z = 1.e^{j\omega}$

$$H(e^{j\omega}) = \frac{1}{6} + \frac{1}{3}e^{-j\omega} + \frac{1}{6}e^{-2j\omega}$$

$$= e^{-j\omega}\left(\frac{1}{6}e^{j\omega} + \frac{1}{3} + \frac{1}{6}e^{-j\omega}\right)$$

$$= e^{-j\omega}\left[\frac{1}{3} + \frac{1}{6}\left[e^{j\omega} + e^{-j\omega}\right]\right]$$

$$= e^{-j\omega}\left[\frac{1}{3} + \frac{1}{6}2\cos\omega\right] \qquad \cos\theta = \frac{e^{j\theta} + e^{-j\theta}}{2}$$

$$= e^{-j\omega}\left[\frac{1}{3} + \frac{1}{3}\cos\omega\right]$$

$$= \underbrace{e^{-j\omega}}_{\text{phase}}\underbrace{\left[\frac{1}{3}(1+\cos\omega)\right]}_{\text{magnitude}}$$

$$= |H(e^{j\omega})| \angle H(e^{j\omega})$$

$$\therefore |H(e^{j\omega})| = \frac{1}{3}(1+\cos\omega) \quad \text{— magnitude response}$$

$$\angle H(e^{j\omega}) = -\omega \quad \rightarrow \text{phase response}$$